

EE6333: Homework 3

Bayesian Estimation

Prof. Sundeep Rangan

Send all submissions to the TA, Mustafa Riza Akdeniz akdeniza@nyu.edu, or submit your answers to NYU Classes.

1. Suppose that $p(x|y) = C(y)e^{-H(x,y)}$ for some functions $C(y)$ and $H(x, y)$.

(a) Show that the MAP estimate of x given y is given by

$$\hat{x} = \arg \min_x H(x, y).$$

(b) Let

$$Z_r := \int x^r \exp(-H(x, y)) dx.$$

Show that the MMSE estimate and its variance is given by

$$\mathbb{E}(x|y) = \frac{Z_1}{Z_0}, \quad \text{var}(x|y) = \frac{Z_2}{Z_0} - \frac{Z_1^2}{Z_0^2}.$$

(c) Show that if $p(x, y) = C(y)e^{-H(x,y)}$, then the formulae in (a) and (b) still hold.

2. Suppose that θ is exponentially distributed with

$$p(\theta) = \frac{1}{\lambda} e^{-\theta/\lambda}, \quad \theta \geq 0,$$

and we have an observation

$$x = \theta + w, \quad w \sim \mathcal{N}(0, \sigma^2),$$

where w is independent of θ .

- (a) What is the ML estimate of θ subject to $\theta \geq 0$?
(b) Find a function $H(\theta, x)$ such that

$$p(\theta|x) = C(x)e^{-H(\theta,x)},$$

for some $C(\theta)$.

- (c) Use the previous problem to compute the MAP estimate. Note that to apply the constraint $\theta \geq 0$, you should separately consider the cases $\theta > 0$ and $\theta = 0$.
(d) Read out a *truncated normal distribution* in

http://en.wikipedia.org/wiki/Truncated_normal_distribution

Show that the posterior density $p(\theta|x)$ is a truncated normal distribution. Use the equations on that page to derive an expression for $\mathbb{E}(\theta|x)$. Your expression will have a Q -function in it.

- (e) Use MATLAB to create a plot with the ML, MAP and MMSE estimate $\hat{\theta}$ for $x \in [-2, 3]$ for $\lambda = 1$ and $\sigma^2 = 0.1$. Plot all three estimators in the same figure and use the legend command to identify the estimators.
3. For a scalar random variable θ , let $\hat{\theta}$ be the estimator that minimizes the expected Bayes risk

$$\hat{\theta} = \arg \min_{\hat{\theta}} \mathbb{E} [L(\theta, \hat{\theta}) | x],$$

for the so-called the L_1 loss function defined by

$$L(\theta, \hat{\theta}) := |\theta - \hat{\theta}|.$$

Thus, $\hat{\theta}$ minimizes the expected absolute difference. Show that if there exists a continuous posterior density $p(\theta|x)$, then $\hat{\theta}$ satisfies

$$P(\theta > \hat{\theta}|x) = P(\theta < \hat{\theta}|x).$$

That is, $\hat{\theta}$ is the value where θ is equally likely to be above and below $\hat{\theta}$, which is the *median* of the density.

Hint: Write

$$J(\hat{\theta}) := \mathbb{E} [L(\theta, \hat{\theta}) | x] = \int_{-\infty}^{\infty} |\theta - \hat{\theta}| p(\theta|x) d\theta.$$

Then, break this integral into two parts: one for $\theta > \hat{\theta}$ and one for $\theta < \hat{\theta}$. Then, take the derivative $J'(\hat{\theta})$ using Leibnitz rule:

http://en.wikipedia.org/wiki/Leibniz_integral_rule

4. A received signal has some power average power level θ . We make N measurements, x_i . Conditional on θ , the power measurements are independent and exponentially distributed with $\mathbb{E}(x_i|\theta) = 1/\theta$.
- (a) What is the ML estimate of θ given the N measurements $\mathbf{x} = (x_1, \dots, x_N)$.
- (b) Look up the Gamma distribution. Suppose that θ has a Gamma prior: $\theta \sim \Gamma(\alpha, \beta)$. Show that the posterior density of θ given $\mathbf{x}$ is also a Gamma with $\theta_{|\mathbf{x}} \sim \Gamma(\alpha', \beta')$ for some α' and β' . What are α' and β' ? This shows that the Gamma distribution is the conjugate prior to the exponential. Feel free to use any facts you can find on wikipedia or any other source. You can state those results without proof.
- (c) Using the results from above, write the posterior mean and variance $\mathbb{E}(\theta|\mathbf{x})$ and $\text{var}(\theta|\mathbf{x})$ in terms of the prior mean $\mu = \mathbb{E}(\theta)$ and variance $\tau = \text{var}(\theta)$ as well as $\mathbf{x}$.
- (d) What is the limit,

$$\lim_{\tau \rightarrow \infty} \mathbb{E}(\theta|\mathbf{x}).$$

How does this compare to the MLE in part (a)?